United States Patent Application Publication	20250280958
Kind Code	A1
Publication Date	September 11, 2025
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BRACE DEVICE AND METHOD

Abstract

A brace device comprising a rest surface for resting a chin, limb, neck or head of a body, a stand surface positioned to abut a surface of the body, a connecting structure coupling the rest surface to the stand surface, a band, and an attachment means provided on the stand surface for securing the band to the stand surface.

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Family ID: **89770460**

Appl. No.: **19/215355**

Filed: **May 22, 2025**

Related U.S. Application Data

parent US continuation 17818111 20220808 parent-grant-document US 12324525 child US 19215355

Publication Classification

Int. Cl.: **A47C7/38** (20060101); **A47C16/00** (20060101)

U.S. Cl.:

CPC **A47C7/383** (20130101); **A47C16/00** (20130101);

Background/Summary

FIELD OF THE DISCLOSURE

[0001] The embodiments described herein relate to bracing devices, and in particular, adjustable bracing devices for the limb, neck, chin, or head of a wearer.

BACKGROUND

[0002] For many passengers the inconveniences of air travel is a small price to pay for a rewarding experience, with several billion passengers flying every year to explore and enjoy new or familiar destinations, comfort during travel can lead to better experiences and prospects in future travel. For travelers, reaching their flight and destination can be an exhaustive process of navigating through the airport, going through security, checking in for their flight, and waiting in traffic and in lines at the airport. Moreover, on their flight or at the airport, passengers often become drowsy, restless, or fatigued and tend to fall asleep while sitting in cramped spaces. Upon waking, passengers may have chin, neck, arm, leg, or other bodily pain from, for example, incorrect posture or being sandwiched in a cramped space. Further, incorrect posture and discomfort from being in a cramped position can startle and wake a sleeping passenger and prevent them from remaining asleep or returning to sleep. Whether traveling for work or leisure, many transportation options exist for travelers, for example, travel by plane, train, metro, boat, trolley, bus, or other vehicle, where many, if not all, travelers must travel in an upright seating position. Moreover, many other factors such as weather, commute time, distance and time of commute, walking and waiting times to reach their transport, and traffic conditions can cause commuters, travelers, or passengers to become drowsy, restless, or fatigued before reaching their destination.

[0003] Incorrect sitting and sleeping posture is very common and much easier and more comfortable to maintain than the proper upright seating or sleeping posture. Most seats on aircraft, trains, buses, and the like, have limited reclining capacity and seating space making it more difficult to maintain a proper sleeping posture in an upright seating position. Further, incorrect sitting and sleeping posture during travel causes discomfort and body pain making it difficult to relax and fall asleep while sitting in upright positions, and there are short term and long term effects to improper sitting and sleeping in upright seating positions. The immediate and short term effect is discomfort, stress on the spine, chin, neck, and head making it difficult for a traveler to be productive at home, at work, or to just enjoy their day. The long term effect from incorrect sitting and sleeping posture can lead to early wear and tear, chronic pain, and surgeries. Therefore, it is important to maintain good posture and comfort in upright seating positions whenever possible.

[0004] To comfort travelers, numerous neck, chin, leg, shoulder, and head support products were developed. For example, neck blankets and neck pillows filled with filler material, air, or made of memory foam were made that are placed on a wearer's shoulders and go around the back of the wearer's neck. These neck blankets and neck pillows provide wearers with a head rest around their neck that supports lying their head on a chair or shoulder to sleep. While the neck blankets and neck pillows can provide seated passengers with some cushioning between the head and seat to allow passengers to lie down, both lack the necessary support to maintain a good posture for sleeping and preventing neck strain or body pain. Furthermore, neck blankets and neck pillows still do not prevent a passenger from waking due to incorrect posture.

[0005] Other examples of passenger support products include pillows, blankets, straps, and padded support structures that secure to the passenger or wedge between the passenger and neighboring chair giving passengers support for sleeping face down on a soft or padded material. While such passenger support products use padding and an alternative sleeping posture in an attempt to mitigate discomfort, because passengers are in a cramped space these products also still fail to provide the necessary support to maintain a good posture for sleeping and to prevent neck strain,

headaches, backaches, or other body pain.
[0006] As another example, various strapping products are available that secure the passenger's body to their seat to maintain a certain posture. Still, as another example, other types of straps and support products are available that attach to a neighboring seat and allow passengers to sleep with their legs stretched out to improve posture and comfort for sleep. While such strapping and support products attempt to mitigate discomfort with alternative sleeping postures, because passengers are in a cramped space it can be very difficult to obtain the correct sleeping posture for various body shapes and sizes, and consequently, for many passengers of varying body shapes and sizes, these products also fail to provide the support needed to maintain a good posture for sleeping and preventing neck strain, headaches, backaches, or other body pain. Moreover, all of these passenger support products fail to provide proper head support and correct posture to prevent the wearer from waking due to involuntary contraction of the muscles of the body which occurs when a person is beginning to fall asleep.

SUMMARY

[0007] The disclosed subject matter relates to a brace device for a chin, limb, neck or head of a body, the brace device including: a rest surface for resting the chin, limb, neck or head of the body, a stand surface positioned to abut a surface of the body, a connecting structure coupling the rest surface to the stand surface, a band, and an attachment means provided on the stand surface for securing the band to the stand surface.
[0008] The disclosed subject matter further relates to a method of supporting a chin, limb, neck or head of a body, the method including the steps of: providing a rest surface for resting chin, the limb, neck or head of the body, providing a stand surface positioned to abut a surface of the body, providing a connecting structure coupling the rest surface to the stand surface, providing a band, and securing one or more portions of the band to the stand surface.
[0009] It is understood that other configurations of the present disclosure will become readily apparent to those skilled in the art from the following detailed description, wherein various configurations of the present disclosure are shown and described by way of illustration. As will be realized, the present disclosure of other different configurations and its several details are capable of modifications in various other respects, all without departing from the subject technology. Accordingly, the drawings and the detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Certain features of the present disclosure are set forth in the appended claims. However, for purpose of explanation, several implementations of the present disclosure are set forth in the following figures.
[0011] FIG. **1**A illustrates a front perspective view of an example brace device in accordance with one or more embodiments of the present disclosure.
[0012] FIG. **1**B illustrates an example configuration of an example brace device in accordance with one or more embodiments of the present disclosure.
[0013] FIG. **2** illustrates an example connecting material used with a brace device in accordance with one or more embodiments of the present disclosure.
[0014] FIG. **3** illustrates a side view of an example brace device in accordance with one or more embodiments of the present disclosure.
[0015] FIG. **4** illustrates a front perspective view of an example brace device being a toy, doll, or stuffed animal, or an example brace device being sheathed in a toy, doll, or stuffed animal cover in accordance with one or more embodiments of the present disclosure.

[0016] FIG. **5**A illustrates an example connecting material connected to an example brace device in accordance with one or more embodiments of the present disclosure.
[0017] FIG. **5**B illustrates an example brace device in accordance with one or more embodiments of the present disclosure.
[0018] FIG. **5**C illustrates another example brace device in accordance with one or more embodiments of the present disclosure.
[0019] FIG. **6**A illustrates an example brace device with one or more movable surfaces in accordance with one or more embodiments of the present disclosure.
[0020] FIG. **6**B illustrates another example brace device with side flaps in accordance with one or more embodiments of the present disclosure.
[0021] FIGS. **6**C-**6**D illustrate another example brace device with one or more movable surfaces in accordance with one or more embodiments of the present disclosure.
[0022] FIG. **7** illustrates an example flow chart showing a method of supporting a chin, limb, neck, or head of a wearer in accordance with one or more embodiments of the present disclosure.
[0023] Embodiments of the present disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like-reference-numerals are used to identify like-elements illustrated in one or more of the figures.

DETAILED DESCRIPTION

[0024] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein may be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale, and the proportions of certain parts have been exaggerated to better illustrate details and features of the present disclosure.
[0025] Various features of the present disclosure will now be described and is not intended to be limited to the embodiments shown herein. Modifications to these features and embodiments will be readily apparent to those skilled in the art, and the principles defined herein may be applied to other embodiments without departing from the scope of the disclosure.
[0026] In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent that the present teachings may be practiced without such details. In other instances, well known methods, procedures, components, and/or circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present teachings.
[0027] Whether traveling for work or leisure, many transportation options exist for travelers, for example, travel by plane, train, metro, boat, trolley, bus, or other vehicle, where many, if not all, travelers must travel in an upright seating position with limited reclining capacity and seating space making it more difficult to maintain a proper sleeping posture. Moreover, incorrect sitting and sleeping posture is very common and much easier and more comfortable to maintain than the proper upright seating or sleeping posture. Further, incorrect sitting and sleeping posture during travel causes discomfort and body pain making it difficult to relax and fall asleep. The immediate and short term effect is discomfort, stress on the spine, chin, neck, and head making it difficult for a traveler to be productive at home, at work, or to just enjoy their day. The long term effect from incorrect sitting and sleeping posture can lead to early wear and tear, chronic pain, and surgeries. To comfort travelers, numerous neck, chin, leg, shoulder, and head support products were developed, however these products fail to provide seated travelers and passengers with proper

support and posture for sitting and sleeping to prevent neck strain, headaches, backaches, or other body pain. Still, some products offer an alternative sleeping posture in an attempt to mitigate discomfort, however, because passengers comprise of various body shapes and sizes, it can be very difficult to obtain the correct sleeping posture for many passengers. Consequently, these products also fail to provide proper support and posture for sleeping to prevent neck strain, headaches, backaches, or other body pain. Moreover, all of these passenger support products fail to provide proper head support and correct posture to prevent the wearer from waking due to involuntary contraction of the muscles of the body which occurs when a person is beginning to fall asleep. Various embodiments of the present disclosure provide a solution to one or more of the above technical problems and others. In one embodiment, a brace device, for providing proper chin, limb, neck, or head support and posture includes a rest surface for resting a chin, limb, neck or head of a wearer, the rest surface is supported by a stand surface and a connecting structure. To facilitate positioning and support the brace device on the body of the wearer, the brace device may include a band and an attachment means provided on the stand surface for securing the band to the stand surface, the band and attachment means being used to secure the brace device to the body of the wearer.

[0028] FIG. **1**A illustrates a front perspective view of an example brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **1**A, brace device **100** may include rest surface **101** configured for resting, for example, a chin, neck, limb, or head of wearer **150** and connecting structure **111** for supporting the position of rest surface **101** as desired by wearer **150**. The brace device **100** may also include stand surface **121** configured for supporting the connecting structure **111** and the position of rest surface **101** as desired by wearer **150**.

[0029] Rest surface **101** includes upper surface **107***a* and lower surface **107***b*. Rest surface **101** may be a chin plate, chin support, chin rest, or the like. In some embodiments, rest surface **101** may be a neck plate, neck support, neck rest, or the like. Still, in some embodiments, rest surface **101** may be a head plate, head support, head rest, or the like. Rest surface **101** may include one or more layers giving it a desired thickness, cushioning or padding, contour, rigidity, or planarity to support the chin, neck, limb, or head of the wearer **150**.

[0030] The upper surface **107***a* may include central portion **106** and one or more lateral portions **103** and **105** attached to the central portion **106** at interfaces **102** and **104**, respectively. In some embodiments, lateral portions **103** and **105** may be planar surfaces that extend away from interfaces **102** and **104**, respectively. Central portion **106** may include one or more non-planar portions positioned laterally along central portion **106** and configured to support the chin, neck, limb, or head of the wearer **150** resting on central portion **106**. In some embodiments, the non-planar portion of central portion **106** includes at least one concave portion or groove **109** formed between interfaces **102** and **104** with lateral portions **103** and **105** on upper surface **107***a* forming planar surfaces extending away from the interfaces **102** and **104**. Referring to FIG. 5C, the groove angle V is the angle of the intersection of the tangents at the ends of the groove **509** at interface **502** and **504**, the groove angle V is configured to be between approximately 5° to 45°. The groove angle V is determined at the surface of the groove **509** in central portion **506** (below padding **510**) and formed between the normal of the rest surface **501** at center line **508** and interface **502** and the normal of the rest surface **501** at center line **508** and interface **504**.

[0031] The stand surface **121** includes front surface **125** and back surface **127** opposite to the front surface **125**. The back surface **127** may be configured to comfortably secure the stand surface **121** to a wearer's body through connecting material **131** and back attachment means positioned on back surface **127**. In some embodiments, front surface **125** may be configured to comfortably secure the stand surface **121** to a wearer's body through connecting material **131** and front attachment means positioned on front surface **125**.

[0032] The connecting material **131** may be, for example, a belt, a strap, a band, a fabric, a scarf, a multi-segmented belt, strap, band, fabric, or any combination thereof. The connecting material **131** includes first end **139**, second end **137**, first wrap portion **133**, second wrap portion **135**. In some embodiments, first and second wrap portions **133**, **135** are configured to wrap around wearer **150** and secure to back surface **127** through back attachment means.

[0033] In some embodiments, only front surface **125** with front attachment means is used to secure the stand surface **121** to the wearer. In some embodiments, only the back surface **127** with front attachment means is used to secure the stand surface **121** to the wearer. In some embodiments, both front surface **125** with front attachment means and back surface **125** with back attachment means are used to secure the stand surface **121** to the wearer. In some embodiments, one end of connecting material **131** may be fixed to stand surface **121** at either the front surface **125** or the back surface **127**, and the opposite end of connecting material **131** may be attached to either front surface **125** with front attachment means or back surface **125** with back attachment means to secure stand surface **121** to the wearer **150**.

[0034] The front surface **125** may include one or more attachment points **125***a*, attachment points **125***a* may be, for example, any one or more protrusions, clips, buckles, hooks, screws, openings, bands, fasteners, magnets, Velcro or Velcro pads, zippers, buttons, or the like, and any combination thereof. Moreover, the attachment points **125***a* may be detachably coupled to front surface **125** through, for example, a removable hook or button that slides or fixes into place on the front surface **125**. In some embodiments, the front surface **125** may include a plurality of attachment points **125***a*. In some embodiments, front surface **125** may include one attachments point **125***a*, a plurality of attachment points **125***a*, or no attachment point. Moreover, each attachment point **125***a* may be the same type or of a different type from one another. In some embodiments, each attachment point **125***a* may be used to hold or secure a device (e.g., mobile device).

[0035] The back surface **127** may include one or more attachment points **127***a*, **127***b*, the attachment points **127***a*, **127***b* may be, for example, any one or more protrusions, clips, buckles, hooks, screws, openings, bands, fasteners, magnets, Velcro or Velcro pads, zippers, buttons, or the like, and any combination thereof. Moreover, the attachment points **127***a*, **127***b* may be detachably coupled to back surface **127** through, for example, a removable hook or button that slides or fixes into place on the back surface **127**. In some embodiments, the back surface **127** may include one or more attachment points **127***a* and **127***b* at each end of back surface **127**. In some embodiments, back surface **127** may include one attachment point **127***a*, two attachment points **127***a*, **127***b*, or no attachment point. Moreover, each attachment point **127***a*, **127***b* may be the same type or of a different type from one another. In some embodiments, the back surface **127** may be configured to have one or more attachments points **127***a* at one end of back surface **127**.

[0036] The lower surface **107***b* of rest surface **101** may be configured to have one or more non-planar, grooved, or concave surface similar to central portion **106**, groove **109**, and planar surfaces extending away from the interfaces **102** and **104** along lateral portions **103** and **105**. In some embodiments, the lower surface **107***b* may be configured to have a planar, curvilinear, or curve surface.

[0037] The connecting structure **111** may be coupled, mounted, fixed, attached, or detachably attached to the lower surface **107***b* of rest surface **101**. In some embodiments, connecting structure **111** may include one or more portions formed between lower surface **107***b* of rest surface **101** and front surface **125** of stand surface **121**. In some embodiments, an end of one portion of the connecting structure **111** may be configured to be coupled, mounted, fixed, attached, or detachably attached to the lower surface **107***b* of rest surface **101**. In some embodiments, connecting structure **111** may include first portion **113** extending at a first angle from lower surface **107***b* and second portion **115** extending at a second angle from an end of the first portion **113**. In some embodiments, the connecting structure **111** may include third portion **117** formed between first portion **113** and second portion **115** and configured to give second portion **115** an offset angle from first portion **113**. In some embodiments, an end of the second portion **115** may be configured to be coupled, mounted, fixed, attached, or detachably attached to the front surface **125** of the stand surface **121**.

The connecting structure **111** may be, for example, substantially cylindrical in shape or rectangular in shape.

[0038] FIG. **1**B illustrates an example configuration of an example brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **1**B, brace device **100** may include first portion **113**, second portion **115**, and third portion **117** each of which extend at different angles from the rest surface **101**. The first portion **113** may be configured to have a predetermined length L.sub.R extending from rest surface **101** and the second portion **115** may be configured to have a predetermined length L.sub.S extending from stand surface **121**. In some embodiments, the third portion **117** may be configured to have a predetermined length L.sub.J extending between the first portion **113** and the second portion **115**. The second portion **115** may extend at a second angle from an end of the first portion **113** and at a third angle from the lower surface **107***b*. In some embodiments, third portion **117** may be formed between first portion **113** and second portion **115**. The third portion **117** may be configured to support better resting angle for resting surface **101** and better chest or body support angle for stand surface **121** by introducing a bend or rotation angle o between first portion **113** and second portion **115**. In some embodiments, the third portion **117** may be a separate material, support, or component from first portion **113** and second portion **115** to facilitate pivoting rest surface **101** and stand surface **121** or to provide one or more rotation angles ϕ between first portion **113** and second portion **115**.

[0039] The rotation angle ϕ, between the first portion **113** and second portion **115** may be configured to be between approximately 0° to 90° making connecting angle C, the angle between first portion **113** and second portion **115**, equal to C=180°−ϕ. For example, the first portion **113** may extend at a normal angle from lower surface **107***b*, at a rotation angle ϕ=0°, the second portion **115** would be parallel to first portion **113** making the connecting angle 180°, at a rotation angle ϕ=10°, the second portion **115** would be rotated inwards at 10°, making the angle between first portion **113** and second portion **115** making the connecting angle 170°.

[0040] The stand angle S, the angle between the second portion **115** and the stand surface **121** may be configured to be between approximately 20° to 60° to provide, for example, support for securing the stand surface **121** to the wearer's chest or body. The stand angle S may be configured with respect to a plane P.sub.W along the front surface **125** of stand surface **121** or tangent to a surface of the wearer's body.

[0041] The rest surface angle R, the angle between the lower surface **107***b* and the first portion **113** may be configured to be between approximately 60° to 120° to provide, for example, a suitable resting surface angle for resting a neck, chin, head, or limb of the wearer. The rest surface angle R may be configured with respect to a plane P.sub.R along the lower surface **107***b* of rest surface **101**.

[0042] In some embodiments, depending on the desired resting surface angle with respect to the chin, head, limb, or neck of the wearer, the rotation angle ϕ, connecting angle C, stand angle S, and rest surface angle R of connecting structure **111** may be configured to provide comfort and support for the chin, neck, limb, or head of the wearer. For example, to provide a desired angle for the second portion **115** (e.g. second portion **115** being parallel to the wearer's neck), the rotation angle ϕ may be configured to be about 20°, the connecting angle C may be configured to be about 160°, the stand angle S may be configured to be about 40°, and rest surface angle R may be configured to be about 90°. Any of the rotation angle ϕ, connecting angle C, stand angle S, and rest surface angle R may be configured with respect to either a plane P.sub.R along the lower surface **107***b* of rest surface **101**, or with respect to a plane P.sub.W along the front surface **125** of stand surface **121** or tangent to a surface of the wearer's body.

[0043] The length L.sub.R of the first portion **113** may be between approximately 2 to 30 centimeters. The length L.sub.S of the second portion **115** may be between approximately 2 to 30 centimeters. The length L.sub.J of the third portion **117** may be between approximately 0.5 centimeters to 10 centimeters. The length L.sub.R of the first portion **113**, length L.sub.S of the second portion **115**, and the length L.sub.J of the third portion **117** may be adjusted as desired to

provide the desired support and a comfortable rest surface **101** angle and position for the wearer **150**.
[0044] The components of the brace device **100** of the present disclosure, the rest surface **101**, connecting structure **111**, and stand surface **121** may be solid, hollow, substantially solid, or substantially hollow and composed of any suitable material, for example metals, plastics, reinforced materials, composites, and the like to ensure easy transport and support and comfort for the wearer. Moreover, the rest surface **101**, connecting structure **111**, and stand surface **121** may be formed as one piece, or a plurality of interconnecting pieces or portions to provide the desired limb, neck, chin or head support. In some embodiments, the brace device **100**, for example, the rest surface **101**, connecting structure **111**, and stand surface **121** may be extruded, printed, molded, or otherwise fabricated together in one or multiple runs as a single piece. Moreover, any one of the brace device components: the upper surface **107***a*, lower surface **107***b*, connecting structure **111**, front surface **125**, back surface **127**, and connecting material **131** may include a padding or cushioning fabric, material, or foam (e.g., memory foam) that covers, sleeves over, or encloses, in part or in whole, said component to obtain, for example, a desired thickness, cushioning or padding, contour, rigidity, or planarity to support the chin, neck, limb, or head of the wearer **150**.
[0045] To provide proper head positioning in an upright seating position that would maintain good posture for sitting or sleeping, the wearer should maintain an appropriate level of neck tilt N.sub.T. The neck tilt N.sub.T is defined as the tilt of the back of the wearer's skull from the plane that crosses the midpoint of the spine. As is well known to one of ordinary skill in the art, a neutral neck provides the least amount of head weight on the spine, that is, the further the wearer's head is tilted from the neutral neck position, the greater the pressure on the spine from the wearer's head. Therefore, it is desirable to maintain the wearer's skull in a neutral neck position, with the back of the skull being vertically aligned with the midpoint of the spine. The brace device of the present disclosure can provide a neck tilt N.sub.T of between approximately 0° to 5° to obtain a neutral neck with minimum head weight of approximately 10-12 lbs. applied to the wearer's spine. Beyond the neutral neck position, the wearer will feel more head weight and stress on their spine, causing the wearer to wake up from their sleep with discomfort and body pain and making it difficult to relax and fall asleep while sitting in upright positions. The distance and angle of rest surface **101** and stand surface **121** with respect to the wearer **150** may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions **113**, **115**, **117** and angles R, ϕ, C and S of the rest surface **101**, the connecting structure **111**, and stand surface **121** for maintaining neutral neck tilt N.sub.T of between approximately 0° to 5° thereby supporting the chin, neck, or head of the wearer **150** in an upright seating position for good sleeping and sitting posture. In some embodiments, the distance and angle of rest surface **101** and stand surface **121** with respect to the wearer **150** may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions **113**, **115**, **117** and angles R, ϕ, C and S of the rest surface **101**, the connecting structure **111**, and stand surface **121** for maintaining proper limb support, angle, positioning, and height relative to the wearer.
[0046] FIG. **2** illustrates an example connecting material used with a brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **2**, wearable material **200** for securing the brace device **100** to the wearer **150** may include connecting material **231** and an attachable collar **241** for cushioning and wearer comfort. The connecting material **231** includes first end **239**, second end **237**, first wrap portion **233**, second wrap portion **235**. In some embodiments, first and second wrap portions **233**, **235** receive the collar **241** for cushioning and wearer comfort. Still, in some embodiments, the first and second wrap portions **233**, **235** may be padded for cushioning and wearer comfort. In some embodiments, the first and second wrap portions **233**, **235** are one piece with collar **249** sheathed, wrapped, attached, or covered around it. In some embodiments, first end **239**, second end **237**, first wrap portion **233**, and second wrap portion **235** of the connecting material **231** may each be detachable from one another and lock or

couple together to form connecting material **231**. For example, first end **239**, second end **237**, first wrap portion **233**, and second wrap portion **235** may each be different colored, textured or designed nylon clip/belts that couple and lock to one another at each end.

[0047] In some embodiments, one end of the connecting material **231** may be fixed to stand surface **121** with the other end detachably coupled to, and freely adjustable on, the stand surface **121** by wearer **150**. The connecting material **231** may include one attachment surface **238** to facilitate securing, or removably attaching, the connecting material **231** to the stand surface **121** through the front or back attachment means. The attachment surface **238** couples, fastens, or secures to the front or back attachment means of the stand surface **121**. For example, to fasten the connecting material **231** to the back surface **127**, the back surface **127** may be configured to have Velcro back attachment means, the attachment surface **238** may be configured to have hook lineal strip and the back surface **127** of the stand surface **121** may be configured to have a loop lineal strip. The hook strip of attachment surface **238** fastens to the loop strip of the back surface **127**. In some embodiments, the attachment surface **238** includes one or more holes to attach to a belt, button, or protrusion attachment means on the stand surface **121**. The attachment surface **238** may, for example, facilitate adjustment and position of the stand surface **121** and rest surface **101** by allowing the wearer to remove and adjust the tension of the connecting material **231** or reposition the stand surface **121** or rest surface **101**.

[0048] In some embodiments, both ends of the connecting material **231** may be detachably coupled to, and freely adjustable on, the stand surface **121** by wearer **150**. In addition to attachment surface **238**, the connecting material **231** may include second attachment surface **236** that couples, fastens, or secures to the front or back attachment means of the stand surface **121**. The second attachment surface **236** may, for example, facilitate ease of access for adjustment and position of the stand surface **121** and rest surface **101** by providing an additional option for the wearer to adjust the tension of the connecting material **231** or reposition the stand surface **121** or rest surface **101**.

[0049] The wearable material **200** may be positioned, connected to, secured, or attached anywhere along the body of wearer **150**. For example, connecting material **231** may comprise of a band, strap, or fabric that, in part or in whole, wraps around, secures, or attaches the stand surface **121** to the body of wearer **150**.

[0050] FIG. **3** illustrates a side view of an example brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **3**, brace device **300** may be secured to the body of wearer **350** using connecting material **331** and stand surface **321**. The tightness or slack of connecting material **331** and the position of stand surface **321** on the wearer's chest may be adjusted to provide desired head support and a comfortable chin rest position on rest surface **301**. Further, the distance and angle of rest surface **301** and stand surface **321** with respect to the wearer **350** may be configured as desired by adjusting the lengths (L.sub.R, L.sub.S and L.sub.J) of first, second and third portions **313**, **315**, **317** and the angles (R, ϕ, C and S) of the rest surface **301**, the connecting structure **311**, and stand surface **321** for supporting the chin, limb, neck, or head of the wearer **350**.

[0051] The rest surface **301** includes right lateral portion **303**, left lateral portion **305**, and central portion **306** on upper surface **307***a*. The rest surface **301** further includes joints or interfaces **302** and **304** between central portion **306** and the right lateral and left lateral portions **303**, **305**. One end of connecting structure **311** is fixed to lower surface **307***b*, the connecting structure **311** extends from the lower surface **307***b* and couples rest surface **301** with stand surface **321**. The connecting structure **311** may include one or more portions to provide the desired head support and a comfortable chin rest position on rest surface **301**. In some embodiments, the connecting structure **311** includes two portions, first portion **311** and second portion **315**. In some embodiments, the connecting structure **311** includes three or more portions, for example, first portion **311**, second portion **315**, and third portion **317**.

[0052] The stand surface **321** includes front surface **325** and back surface **327**. The front surface

325 may include front attachment means having one or more attachment points **325***a* that secure the wearable material **200** or connecting material **331** to the front surface **325**. The back surface **327** may include back attachment means having one or more attachment points **327***a*, **327***b* that secure the wearable material **200** or connecting material **331** to the back surface **327**.

[0053] The connecting material **331** may include detachable collar **341** for cushioning and wearer comfort. The connecting material **231** includes first end **339**, second end **337**, first wrap portion **333**, second wrap portion **335**. In some embodiments, first and second wrap portions **333**, **335** receive the collar **341** for cushioning and wearer comfort. Still, in some embodiments, the first and second wrap portions **333**, **335** may be padded for cushioning and wearer comfort.

[0054] The distance and angle of rest surface **301** and stand surface **321** with respect to the wearer **350** may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions **313**, **315**, **317** and angles R, ϕ, C and S of the rest surface **301**, the connecting structure **311**, and stand surface **321** for maintaining neutral neck tilt N.sub.T of between approximately 0° to 5° thereby supporting the neck, head, or chin of the wearer **350** in an upright seating position for good sleeping and sitting posture. In some embodiments, the distance and angle of rest surface **301** and stand surface **321** with respect to the wearer **350** may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions **313**, **315**, **317** and angles R, ϕ, C and S of the rest surface **301**, the connecting structure **311**, and stand surface **321** for maintaining proper limb support, angle, positioning, and height relative to the wearer.

[0055] FIG. **4** illustrates a front perspective view of an example brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **4**, the rest surface **401**, connecting structure **411**, and the stand surface **421** of brace device **400** may each be covered by one or more parts of a toy, doll, or stuffed animal (hereinafter “stuffed animal **460**”). In some embodiments, the stuffed animal **460** may include components of brace device **400**. The rest surface **401** may form the head of stuffed animal **460**, connecting structure **411** may form the neck, upper torso or abdomen of stuffed animal **460**, and the stand surface **421** may form the limbs, back, or lower torso or abdomen of stuffed animal **460**. For example, the head of stuffed animal **460** may form upper surface **407***a* and lower surface **407***b* of rest surface **401**. The torso of stuffed animal **460** may include front surface **425** and back surface **427**. The front surface **425** may include one or more attachment points **425***a* (e.g. limbs, legs, or torso of the stuffed animal **460**) and an attachment means for securing stand surface **421** to the wearer **450**. Further, the back surface **427** may include one or more attachment points **427***a* and **427***b* and an attachment means for securing stand surface **421** to the wearer through connecting material **431**. In central portion **406** of rest surface **401** includes non-planar portion **409** formed in upper surface **407***a* to provide the desired head support and a comfortable chin rest position on rest surface **401**.

[0056] The connecting material **431** may include detachable collar **241** for cushioning and wearer comfort. The connecting material **431** includes first end **439**, second end **437**, first wrap portion **433**, second wrap portion **435**. In some embodiments, first and second wrap portions **433**, **435** receive the collar **241** for cushioning and wearer comfort. Still, in some embodiments, the first and second wrap portions **433**, **435** may be padded for cushioning and wearer comfort.

[0057] The distance and angle of rest surface **401** and stand surface **421** with respect to the wearer **450** may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions of the connecting structure **411** and angles R, ϕ, C and S of the rest surface **401**, the connecting structure **411**, and stand surface **421** for maintaining neutral neck tilt N.sub.T of between approximately 0° to 5° thereby supporting the neck, head, or chin of the wearer **450** in an upright seating position for good sleeping and sitting posture. In some embodiments, the brace device **400** is configured as desired to maintain neutral neck tilt NT when sheathed or covered by a toy, doll, or stuffed animal. In some embodiments, the brace device **400** may be a toy, doll, or stuffed animal with predetermined lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions of the connecting structure **411** and predetermined angles R, ϕ, C and S of the rest surface

401, the connecting structure **411**, and stand surface **421** for maintaining neutral neck tilt NT of between approximately 0° to 5° thereby supporting the neck, head, or chin of the wearer **350** in an upright seating position for good sleeping and sitting posture. In some embodiments, the distance and angle of rest surface **401** and stand surface **421** with respect to the wearer **450** may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions of the connecting structure **411** and angles R, ϕ, C and S of the rest surface **401**, the connecting structure **411**, and stand surface **421** for maintaining proper limb support, angle, positioning, and height relative to the wearer.

[0058] FIG. **5**A illustrates an example connecting material connected to an example brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **5**A, rest surface **501** may include one or more attachment points **501***a*, **501***b*, and **501***c* either on upper surface **507***a* or lower surface **507***b*, or both, for attaching a padding or cushioning fabric, material, or foam (e.g., memory foam) that covers the rest surface **501**, in part or in whole, to obtain, for example, a desired thickness, cushioning or padding, contour, rigidity, or planarity to support the chin, neck, limb, or head of the wearer **150**. The attachment points **501***a*, **501***b*, and **501***c* may include one or more attachment means, for example, any one or more protrusions, clips, buckles, hooks, screws, openings, bands, fasteners, magnets, Velcro or Velcro pads, zippers, buttons, or the like, and any combination thereof for securing a padding or cushioning fabric, material, or foam (e.g., memory foam) to the brace device **500**.

[0059] In some embodiments, the padding cover may include one or more parts of a toy, a doll, or a stuffed animal. In some embodiments, a head of stuffed animal **460** may enclose, cover, or sleeve over the rest surface **501**, central portion **506**, groove **509**, and left lateral and right lateral portions **503** and **505** to support the chin, neck, limb, or head of the wearer **150**. Moreover, any one of attachment points **501***a*, **501***b*, and **501***c* may be positioned on interface **502** or **504** to support one or more padding covers (e.g., the head of stuffed animal **460**). In some embodiments, a plurality of attachment points **501***a*, **501***b*, and **501***c* are positioned on the brace device **500**. For example, one or more attachments points **501***a*, **501***b*, and **501***c* may be positioned on connecting structure **511**, first portion **513**, second portion **515**, stand surface **521**, front surface **525**, and back surface **527** to support one or more padding covers (e.g., the body, or parts of, a toy, doll, or stuffed animal).

[0060] The brace device **500** includes connecting material **531** fixed at the first end **537** to back attachment point **527***a* of back surface **527**. In this example, the back attachment means may be, for example, glue, epoxy, rivet that secures the first end **537** to back surface **527**. In some embodiments, the back attachment means may be by stamping, pressing, gluing, or otherwise adhering a portion of the first end **537** to the back attachment point **527***a*. In some embodiments, attachment point **527***a* may be one or more structures that is positioned over first end **537** and stamped, pressed, adhered or otherwise secured to the back surface **527**. The brace device **500** further includes another attachment point **527***b* that detachably couples or secures attachment surface **238** on the second end **539** of connecting material **531**. In some embodiments, the back attachment means may further include stamping, pressing, gluing, or otherwise adhering a portion of the second end **539** to the back attachment point **527***b*. The connecting material **531** may then be stretched or pulled over the body of the wearer, first end **537**, second end **539**, first and second wrap portions **533** and **535** may each be nylon clip/belts that detachably coupled to one another, and the collar **541** may removably attached to connecting material **531**. In some embodiments, attachment point **527***b* may be one or more structures that is positioned over first end **539** and stamped, pressed, adhered or otherwise secured to the back surface **527**. In some embodiments, padding or collar **541** may be fixed to the connecting material **531** between first and second wrap portions **533** and **535**.

[0061] FIG. **5**B illustrates an example brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **5**B, the rest surface **501** may exclude left and right lateral portions **503** and **505** with the central portion **506** and groove **509** extending between interfaces **502** and **504** which also form the edges of the upper surface **507***a* and lower surface **507***b*. Moreover, the connecting structure **511** may form continuous curved structure that attaches stand surface **521** to rest surface **501**.

[0062] FIG. **5**C illustrates another example brace device in accordance with one or more embodiments of the present disclosure. As shown in FIG. **5**C, the rest surface **501** may include a memory foam or padding **510** attached over part or all of the central portion **506** and groove **509**. Moreover, increasing the thickness of padding **501** reduces the groove angle V to provide the desired contour or cushioning to support the chin, neck, limb, or head of the wearer **150**. The groove angle V is the angle of the intersection of the tangents at the ends of the groove **509** at interface **502** and **504**, the groove angle V is configured to be between approximately 5° to 45°. The groove angle V is determined at the surface of the groove **509** in central portion **506** (below padding **510**) and formed between the normal of the rest surface **501** at center line **508** and interface **502** and the normal of the rest surface **501** at center line **508** and interface **504**. Moreover, the connecting structure **511** may form a single rigid or continuous structure, for example substantially cylindrical or substantially rectangular in shape, that attaches stand surface **521** to rest surface **501**.

[0063] Referring to FIGS. **5**A-**5**C, and as described above, the distance and angle of rest surface **501** and stand surface **521** with respect to the wearer may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions of the connecting structure **511** and angles R, ϕ, C and S of the rest surface **501**, the connecting structure **511**, and stand surface **521** for maintaining neutral neck tilt N.sub.T of between approximately 0° to 5° thereby supporting the neck, head, or chin of the wearer in an upright seating position for good sleeping and sitting posture. In some embodiments, the distance and angle of rest surface **101** and stand surface **121** with respect to the wearer may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions of connecting structure **511** and angles R, ϕ, C and S of the rest surface **501**, the connecting structure **511**, and stand surface **521** for maintaining proper limb support, angle, positioning, and height relative to the wearer.

[0064] FIG. **6**A illustrates an example brace device with one or more movable surfaces in accordance with one or more embodiments of the present disclosure. As shown in FIG. **6**A, brace device **600** includes rotatable flaps **671** and **673** connected to pivoting structures **602***a* and **604***a*, respectively. The pivot structure **602***a* and **604***a* may be a barrel attached at interface **602** and **604**, respectively. The flap **671** may form the right lateral portion **603** and flap **673** may form the left lateral portion **605** on rest surface **601**. The upper surface **607***a* includes central portion **606** and groove **609**. In some embodiments, lower surface **607***b* may be configured to have one or more non-planar, grooved, or concave surfaces similar to groove **609** of central portion **606** and planar surfaces extending away from the interfaces **602** and **604** along right and left lateral portions **603** and **605**. In some embodiments, the lower surface **607***b* may be configured to have a planar, curvilinear, or curve surface. The rest surface **601** being fixed to stand surface **621** through connecting structure **611**.

[0065] FIG. **6**B illustrates another example brace device with side flaps in accordance with one or more embodiments of the present disclosure. As shown in FIG. **6**A, brace device **600** includes fixed flaps **671** and **673** connected either at the ends of right and left lateral portions **603** and **605**, or at the outside edges of right and left lateral portions **603** and **605**. In some embodiments, side flaps **671**, **673** are mounted, fixed, or otherwise attached adjacent to or along the outer edge of rest surface **601**. As an example, the wearer may use side flaps **671**, **673** to vertically position their chin, limb, neck or head at a desired angle above rest surface **601**.

[0066] FIGS. **6**C-**6**D illustrate another example brace device with one or more movable surfaces in accordance with one or more embodiments of the present disclosure. As shown in FIGS. **6**C-**6**D, brace device **600** includes rotatable flaps **671** and **673** connected to pivoting structures **602***a* and **604***a*, respectively. The pivot structure **602***a* and **604***a* may be a barrel attached at interface **602** and

604, respectively. The flaps **671** and **673** may folded down over right lateral portion **603** and flap **673** for storage or transport. The upper surface **607***a* includes central portion **606**, groove **609**. In some embodiments, lower surface **607***b* may be configured to have one or more non-planar, grooved, or concave surface similar to groove **609** of central portion **606** and planar surfaces extending away from the interfaces **602** and **604** along right and left lateral portions **603** and **605**. In some embodiments, the lower surface **607***b* may be configured to have a planar, curvilinear, or curve surface. In some embodiments, one or more flaps **671**, **673** are movable, for example, pivotable, rotatable, etc., and are mounted, fixed, or otherwise attached adjacent to or along the outer edge of rest surface **601**. As an example, the wearer may use movable flaps **671**, **673** to position each flap **671**, **673** at a desired angle to vertically position their chin, limb, neck or head at a desired angle above rest surface **601**.

[0067] Referring to FIGS. **6**A-**6**D, and as described above, the distance and angle of rest surface **601** and stand surface **621** with respect to the wearer may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.j of first, second and third portions of the connecting structure **611** and angles R, ϕ, C and S of the rest surface **601**, the connecting structure **611**, and stand surface **621** for maintaining neutral neck tilt N.sub.T of between approximately 0° to 5° thereby supporting the neck, head, or chin of the wearer in an upright seating position for good sleeping and sitting posture. In some embodiments, the distance and angle of rest surface **601** and stand surface **621** with respect to the wearer may be configured by adjusting lengths L.sub.R, L.sub.S and L.sub.J of first, second and third portions of connecting structure **611** and angles R, ϕ, C and S of the rest surface **601**, the connecting structure **611**, and stand surface **621** for maintaining proper limb support, angle, positioning, and height relative to the wearer.

[0068] FIG. **7** illustrates an example flow chart showing a method of supporting a limb, neck, chin, or head of a wearer of a bracing device in accordance with one or more embodiments of the present disclosure. These exemplary methods are provided by way of example, as there are a variety of ways to carry out these methods. Each block shown in FIG. **7** represents one or more processes, methods or subroutines, carried out in the exemplary method. FIGS. **1**A-**6**D show example embodiments of carrying out the method of FIG. **7** for supporting a limb, neck, chin, or head of a wearer of a bracing device. Each block shown in FIG. **7** represents one or more processes, methods or subroutines, carried out in the exemplary method. The exemplary method may begin at block **701**. Method **700** may be used independently or in combination with other methods or process for supporting a limb, neck, chin, or head of a wearer of a bracing device. For explanatory purposes, the example process **700** is described herein with reference to the brace device of FIGS. **1**A-**6**D. Further for explanatory purposes, the blocks of the example process **700** are described herein as occurring in serial, or linearly. However, multiple blocks of the example process **700** may occur in parallel. In addition, the blocks of the example process **700** may be performed a different order than the order shown and/or one or more of the blocks of the example process **700** may not be performed. Further, any or all blocks of example process **700** may further be combined and done in parallel, in order, or out of order.

[0069] In FIG. **7**, the exemplary method **700** of implementing a support device for supporting a limb, neck, chin, or head of a wearer is shown. Method **700** begins at block **701**. In block **703**, the method includes providing a rest surface, a stand surface, a connecting structure, and a band of a support device, the connecting structure coupling the rest surface to the stand surface.

[0070] In block **705**, the method includes securing one or more portions of the band to the stand surface. In some embodiments, securing one or more portions of the band further comprises placing the band around the body, wherein the band extends between the stand surface and a first portion of the body and goes around a second portion of the body.

[0071] In block **707**, the method includes placing a chin, limb, neck or head of a body on the rest surface. In some embodiments, the rest surface may include a non-planar portion, and supporting the limb, neck, chin, or head may further comprise placing the chin, limb, neck or head of the body on the non-planar portion of the rest surface. The non-planar portion may be concave for receiving the chin, limb, neck or head or convex for supporting the chin, limb, neck or head.

[0072] In block **709**, the method includes placing the band around the body. In some embodiments, a collar or a padding may be placed on the band to provide better support or comfort for the wearer.

[0073] In block **711**, the method includes repositioning the stand surface by detaching at least one portion of the one or more portions of the band from the stand surface.

[0074] In block **713**, the method includes adjusting a flap mounted adjacent to or along an outer edge of the rest surface to position the chin, limb, neck or head of the body at a desired angle from the rest surface. In some embodiments, the rest surface includes one or more flaps attached at one or both ends of the groove on the upper surface of the support device. In some embodiments, at least one of the one or more flaps is pivotable from the outer edge of the rest surface, and placing the chin, limb, neck, or head on the rest surface further comprises adjusting the pivotable flap to position the chin, limb, neck or head of the body at a desired angle from the rest surface. The method ending in block **715**.

[0075] The term “within a proximity”, “a vicinity”, “within a vicinity”, “within a predetermined distance”, “predetermined width”, “predetermined height”, “predetermined length” and the like may be defined between about 0.1 centimeter and about 0.5 meters. The term “coupled” is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The connection may be such that the objects are permanently connected or releasably connected. The term “substantially” is defined to be essentially conforming to the particular dimension, shape, or other feature that the term modifies, such that the component need not be exact. For example, “substantially cylindrical” means that the object resembles a cylinder, but may have one or more deviations from a true cylinder. The term “comprising,” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series and the like.

[0076] The term “a predefined” or “a predetermined” when referring to length, width, height, or distances may be defined as between about 0.1 centimeter and about 0.5 meters.

[0077] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the present disclosure, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the present disclosure or that such disclosure applies to all configurations of the present disclosure. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0078] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” or as an “example” is not necessarily to be construed as preferred or advantageous over other embodiments. Furthermore, to the extent that the term “include”, “have”, or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

[0079] All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

[0080] The previous description of the disclosed embodiments is provided to enable a person skilled in the art to make or use the disclosed embodiments. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the principles defined herein may be applied to other embodiments without departing from the scope of the disclosure. Thus, the present disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope possible consistent with the principles and novel features as defined by the following claims.

[0081] The embodiments shown and described above are only examples. Many details are often found in the art such as the other features of an image device. Therefore, many such details are neither shown nor described. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, especially in matters of shape, size, and arrangement of the parts within the principles of the present disclosure, up to and including the full extent established by the broad general meaning of the terms used in the claims. It will therefore be appreciated that the embodiments described above may be modified within the scope of the claims.

Claims

1. A brace device, comprising: a rest surface for resting at least one of a chin, limb, neck and head of a body; a stand surface positioned to abut a surface of the body; a connecting structure, the connecting structure configured as a single piece coupling the rest surface to the stand surface and comprising a bend positioned away from the distal ends of the connecting structure; a distal end of the connecting structure coupled to the rest surface and an other distal end of the connecting structure coupled to the stand surface; and an attachment means provided on the connecting structure for securing a connecting material to the connecting structure.

2. The brace device of claim 1, wherein the rest surface includes a non-planar portion for supporting at least one of the chin, limb, neck and head.

3. The brace device of claim 2, wherein the non-planar portion is concave.

4. The brace device of claim 1, wherein the connecting structure comprises one or more openings, clips, screws, belts, buckles, hooks, or any combination thereof.

5. The brace device of claim 1, wherein the connecting structure extends at an angle from the stand surface.

6. The brace device of claim 1, further comprising a band, wherein the band includes a collar or a padding to support the chin, limb, neck or head.

7. The brace device of claim 1, further comprising a band, wherein the band extends between the connecting structure and a first portion of the body and goes around a second portion of the body.

8. The brace device of claim 1, further comprising a band, wherein the band is detachable from the attachment means to facilitate repositioning of the rest surface.

9. The brace device of claim 1, wherein the rest surface further comprises one or more flaps mounted adjacent to an outer edge of the rest surface.

10. The brace device of claim 9, wherein at least one of the one or more flaps is pivotably connected to the rest surface to vertically position at least one of the chin, limb, neck and head above the rest surface.

11. The brace device of claim 1, wherein the rest surface, the connecting structure, and the stand surface together form one or more parts of a toy, a doll, or a stuffed animal.

12. A method of supporting at least one of a chin, limb, neck and head of a body, comprising: providing a rest surface for resting the chin, limb, neck or head of the body; providing a stand

surface positioned to abut a surface of the body; coupling, by a connecting structure, the rest surface to the stand surface, the connecting structure configured as a single piece coupling the rest surface to the stand surface and comprising a bend positioned away from the distal ends of the connecting structure; configuring a distal end of the connecting structure coupled to the rest surface and an other distal end of the connecting structure coupled to the stand surface; and securing one or more portions of a connecting material to the connecting structure.
13. The method of claim 12, wherein the rest surface includes a non-planar portion, and supporting at least one of the chin, limb, neck and head, further comprises placing the chin, limb, neck or head on the non-planar portion of the rest surface.
14. The method of claim 13, wherein the non-planar portion is concave for receiving the chin, limb, neck or head.
15. The method of claim 13, wherein the non-planar portion is convex for supporting the chin, limb, neck or head.
16. The method of claim 12, further comprising placing the connecting material around the body, wherein the connecting material extends between the connecting structure and a first portion of the body and goes around a second portion of the body.
17. The method of claim 12, further comprising repositioning the stand surface on the body by detaching at least one portion of the one or more portions of the connecting material from the connecting structure.
18. The method of claim 12, wherein the rest surface includes one or more flaps mounted adjacent to an outer edge of the rest surface, and placing at least one of the chin, limb, neck and head on the rest surface comprises positioning the chin, limb, neck or head between the rest surface and the one or more flaps.
19. The method of claim 18, wherein at least one of the one or more flaps is pivotably attached to the rest surface, and placing at least one of the chin, limb, neck, and head on the rest surface further comprises adjusting the at least one pivotable flap to vertically position the chin, limb, neck or head at a desired angle from the rest surface.
20. A neck brace device for maintaining a neck tilt for the wearer, comprising: a rest surface for resting at least one of a chin, limb, neck and head of a body; a stand surface positioned to abut a surface of the body; a connecting structure, the connecting structure configured as a single piece coupling the rest surface to the stand surface and comprising a bend positioned away from the distal ends of the connecting structure; a distal end of the connecting structure coupled to the rest surface and an other distal end of the connecting structure coupled to the stand surface; and an attachment means provided on at least one of the stand surface and the connecting structure for securing a connecting material therefrom; wherein the attachment means, when secured by the connecting material, maintains a neck tilt of between 0 degrees to 5 degrees for the wearer.